

5.1 Práctica: Diseño de un controlador para un sistema de segundo orden.

Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t))$$

$$L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t)) = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R I_1(s) + L s (I_1(s) - I_2(s)) + R(I_1(s) - I_2(s))$$

$$L s (I_1(s) - I_2(s)) + R(I_1(s) - I_2(s)) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

$$V_e(s) = R I_1(s) + L s I_1(s) - L s I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (L s + 2R) I_1(s) - (L s + R) I_2(s)$$

$$I_1(s) = \frac{1}{L s + R} \left(L s + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs} \right) I_2(s)$$

$$I_1(s) = \frac{1}{Ls+R} \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$I_1(s) = \frac{\frac{CsLs+1+3RCs}{Cs}}{Ls+R} I_2(s)$$

$$I_1(s) = \frac{CLs^2 + 3RCs + 1}{CLs^2 + RCs} I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs} \right) I_2(s)$$

$$V_s(s) = \frac{CsR + 1}{Cs} I_2(s)$$

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3RCs + 1}{CLs^2 + RCs} I_2(s) - (Ls + R) I_2(s)$$

$$V_e(s) = \left[(Ls + 2R) \frac{CLs^2 + 3RCs + 1}{CLs^2 + RCs} - (Ls + R) \right] I_2(s)$$

$$V_e(s) = \frac{(Ls + 2R)(CLs^2 + 3RCs + 1) - (CLs^2 + RCs)(Ls + R)}{CLs^2 + RCs} I_2(s)$$

$$V_e(s) = \frac{\cancel{CL^2s^2} + 3\cancel{CLR}s^2 + \cancel{Ls} + 2\cancel{CLR}s^2 + 6\cancel{CR^2} + 2R - \cancel{CL^2s^2} + \cancel{CLR}s^2}{CLs^2 + RCs} I_2(s)$$

$$= \frac{-\cancel{CRL}s^2 - \cancel{CR^2}s}{CLs^2 + RCs} I_2(s)$$

$$V_e(s) = \frac{3CLR s^2 + (5CR^2 + L)s + 2R}{CLs^2 + RCs} I_2(s)$$

Función de transferencia.

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs+1}{Cs} I_2(s)}{\frac{3CLR s^2 + (SCR^2+L)s + 2R}{CLs^2 + CRs} I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{C^2 L R s^3 + CLs^2 + C^2 R^2 s^2 + CRs}{3C^2 L R s^3 + Cs^2 (SCR^2 + L) + 2CRs}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\cancel{Cs} (CLR s^2 + Ls + CR^2 + R)}{\cancel{Cs} (3CLR s^2 + s(SCR^2 + L) + 2R)}$$

$$\boxed{\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + (L + CR^2)s + R}{3CLR s^2 + (SCR^2 + L)s + 2R}}$$

Error en estado estacionario.

$$e(s) = \lim_{s \rightarrow 0} s V_e(s) \left(1 - \frac{V_s(s)}{V_e(s)} \right)$$

$$e(s) = 1 - \frac{R}{2R}$$

$$\boxed{e(s) = 1/2}$$

Estabilidad en lazo abierto.

$$R = 1k\Omega$$

$$L = 47 \mu H$$

$$C = 470 \mu F$$

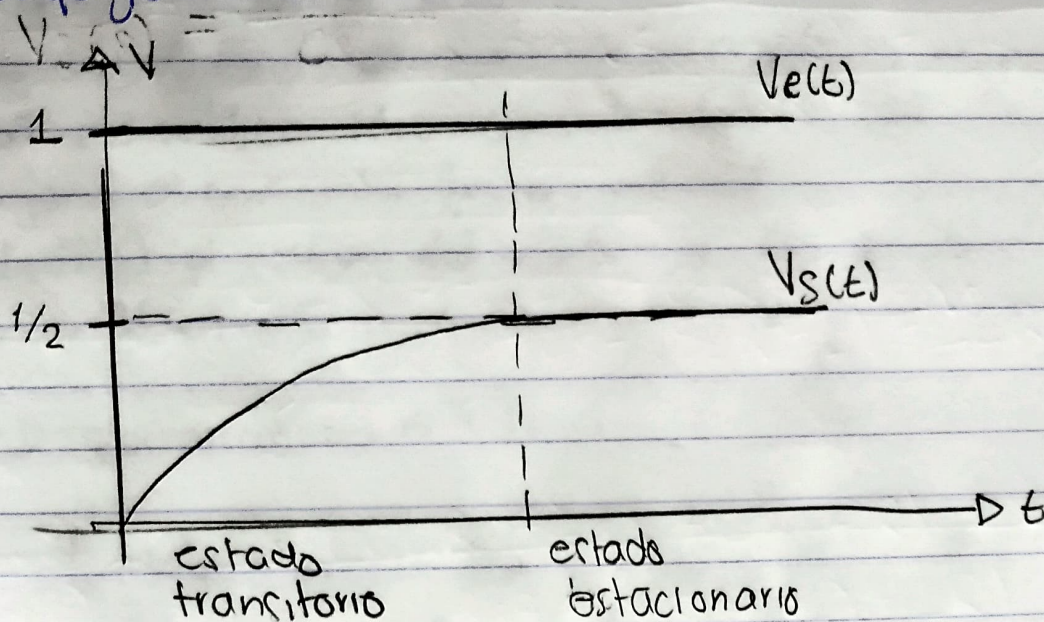
$$3CLR s^2 + (SCR^2 + L)s + 2R = 0$$

$$s_1 = -.85106$$

$$s_2 = -35460992.77$$

El sistema es estable con una respuesta sobreamortiguada por sus polos reales negativos.

Bosquejo de la función



Modelo de ecuaciones integro-diferenciales

El modelo se expresa despejando las variables dependientes, es decir las corrientes

$$L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t)) = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$L \frac{d(i_1(t) - i_2(t))}{dt} + R i_1(t) + R i_2(t) = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$L \frac{d(i_1(t) - i_2(t))}{dt} + R i_1(t) = 2R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d(i_1(t) - i_2(t))}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{2R}$$

$$V_e(t) = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t))$$

$$V_e(t) = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R(i_1(t) - i_2(t)) + R i_2(t)$$

$$V_e(t) = L \frac{d(i_1(t) - i_2(t))}{dt} + R i_2(t) = 2R i_1(t)$$

$$\uparrow i_1(t) = \left[V_e(t) - L \frac{d(i_1(t) - i_2(t))}{dt} + R i_2(t) \right] \frac{1}{2R}$$

$$\uparrow V_0(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$